



# Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

## 1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

|                                                                               |                                                                                                                  |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Imię i nazwisko / Name, surname                                               | Grzegorz Piotrowski                                                                                              |
| Dyscyplina kształcenia / Discipline of education                              | informatyka techniczna i telekomunikacja / information and communication technology                              |
| Wydział / Faculty                                                             | W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology                   |
| Katedra / Department                                                          | K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence                                       |
| Data rozpoczęcia kształcenia / Date of commencement of the doctoral education | 01.10.2025                                                                                                       |
| Nr ORCID                                                                      | 0009-0004-6013-0461                                                                                              |
| Promotor/ promotorzy / Supervisor(s)                                          | dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012 |
| Promotor pomocniczy / Assistant supervisor                                    | —                                                                                                                |

## 2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

Metody uczenia reprezentacji dla generalizacji modeli uczenia głębokiego między symulacją a rzeczywistością w fizycznej sztucznej inteligencji / Representation learning methods for simulation-to-reality generalization of deep learning models in physical AI

## 3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,



- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

*A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:*

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

| <b>Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)</b> | <b>Zwięzły opis zadania / Brief description of the task</b>                                                                                                        |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semestr 1                                                                                     | Doctoral School courses; reading the literature and considering possible research directions                                                                       |
| Semestr 2                                                                                     | Doctoral School courses; literature review; choice of the dissertation topic with the supervisor; preparing the Individual Research Plan                           |
| Semestr 3                                                                                     | Preparing the research environment (digital twins, simulators, public datasets); initial experiments on harmful reconstruction errors (RQ1)                        |
| Semestr 4                                                                                     | Further experiments on RQ1; preparing and submitting the first publication to a robotics conference or journal (e.g. ICRA or RA-L); first experiments on RQ2       |
| Semestr 5                                                                                     | Experiments on localizing the gap inside models and aligning representations (RQ2); preparing a scientific publication                                             |
| Semestr 6                                                                                     | Experiments on selecting real data (RQ3) and on transfer to unseen scenes and between tasks (RQ4); preparing a publication; tests on a real robot if access allows |
| Semestr 7                                                                                     | Final experiments and submission of a paper to a conference or journal; first draft of the dissertation                                                            |
| Semestr 8                                                                                     | Completing, revising and submitting the doctoral dissertation                                                                                                      |

#### **4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission**



*Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.*

## September 2029

*(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /*

*The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)*

## 5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

*Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).*

Deep learning has made great progress in computer vision and language processing, largely thanks to vast amounts of data collected from the web. Physical AI, that is, AI in robots and other systems that perceive and act in the real world, cannot rely on such data. Collecting interaction data with real robots is slow, expensive and sometimes unsafe, so robot models are commonly trained in simulation. Recently, digital twins, that is, simulations built from real data, have become a promising alternative to hand-crafted simulators. Neural scene reconstruction methods such as 3D Gaussian Splatting turn a short recording of a real place into a photorealistic copy, in which a model can be trained and then transferred back to reality.

However, a model that performs well in its digital twin often fails in reality, and even more often in places it has never seen. No twin reproduces the world exactly. Appearance, geometry, lighting and physics are only approximated, and reconstruction errors differ between regions of a scene. The resulting drop in performance is called the simulation-to-reality gap. It is a case of learning under distribution shift and one of the main obstacles to physical AI.

The methods used today address it only in part. Domain randomization varies a few global simulator parameters, regardless of where the twin is actually wrong. Invariant representations may discard information that the task needs. The few real data collected for adaptation usually correct either the model or the simulation, but not both. Most methods are also evaluated on a single task and a few scenes, so it is not known which properties of a twin and of the learned representations decide whether a model will work in reality.

My research aims to reduce this gap with methods that analyse and align the representations learned by the models, applied where the gap arises. I want to find out which reconstruction errors of a twin actually harm a model, at which stage inside the model task information is lost on real inputs, and which small set of real data best corrects both the twin and the model. I will also check whether the improvements carry over to unseen scenes and from one task to another. I will study two tasks of equal weight, robot navigation and robotic manipulation, and measure all results on scenes that were not used for training.

The results could be applied in service, logistics and assistive robotics, where a robot could be trained in a digital copy of a new warehouse, hospital or home and then work there reliably after little real-world adaptation. They also concern autonomous driving and inspection robots, whose models are likewise trained on simulated or reconstructed data.



## 6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**

*Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).*

Deep learning models for robot navigation and manipulation need more experience than real robots can provide, so they are usually trained in simulation. A study on navigation showed that performance in simulation can poorly predict performance in reality unless the simulator is carefully tuned [1]. The simulation-to-reality gap is a case of learning under distribution shift. Domain adaptation theory bounds the error of a model on the target domain by its error on the source domain, the discrepancy between the two distributions and the joint error of the best single model on both domains [2]. I treat this bound only as motivation. It suggests making the training distribution cover reality, making the model insensitive to the remaining discrepancy and, since invariance does not target the joint error and can even increase it, correcting the simulation towards reality, which can reduce this error directly.

A widely used approach is domain randomization, which varies simulator parameters such as textures so that reality appears to the model as one more variation [3]. Overly wide randomization leads to conservative behaviour, so later works shape its distribution by fitting it to a few real rollouts [4] or by maximizing its entropy while preserving task success [5]. These methods tune a few global parameters of a synthetic simulator. They do not address errors that vary across a scene, which are typical of twins reconstructed from real data.

Another approach is to make the model invariant to the domain. Domain-adversarial training makes the features of the two domains indistinguishable to a domain classifier [6]. In robotics, aligning the joint distributions of observations and actions in simulated and real data improved co-trained policies, that is, models that map observations to actions, in reality [7]. However, invariance with low source error does not guarantee transfer. It can increase the joint error term when the label distributions, here the actions, differ between domains [8], and full invariance may discard task information. A recent analysis showed that effective co-training on simulated and real data aligns the two domains while keeping them distinguishable [9], so domain separability alone does not show where the gap harms the task. Invariance is also usually enforced at locations chosen in advance, such as the input, the final features or fixed layers, rather than where the gap is measured to arise.

Digital twins built from real data are a more recent direction. 3D Gaussian Splatting [10] reconstructs photorealistic scenes from multi-view photographs and renders them in real time. In manipulation, twins have been used to train policies from a scan of the target scene [11], to transfer image-based policies to reality without real-world demonstrations [12] and to guide real-world reinforcement learning from a phone capture [13]. The SIMPLER study [14] showed with paired simulated and real evaluations that scenes matched to real images and controllers of two robot setups track the real-world performance of the same policies. In navigation, policies have been fine-tuned in twins built from a phone capture of a room [15] and trained in twins built from monocular urban videos [16]. The effect of simulation factors has been studied mainly globally, for example randomization, photorealism and physics realism in real manipulation [17]. For whole workspaces, the similarity of pretrained features between rendered and real views predicted the agreement of simulated and real performance better than LPIPS or PSNR [18]. In my view, it remains open for both tasks which reconstruction errors that vary across a scene harm generalization, and how training variation should follow the reconstruction error and task relevance of each region.



It is also unclear where inside a model the gap arises. Probing, that is, reading out information from hidden states with simple classifiers, showed that fine-tuning for actions degrades the visual representations of vision-language-action models [19]. Separately, fine-tuning only selected layers showed that the best layers to adapt depend on the type of shift [20]. To my knowledge, in models trained in twins, the stage at which task information available in simulation stops being recoverable from real inputs has not been located.

The remaining gap is usually reduced with a small amount of real data. Co-training with a small real set yields large gains [21], and active system identification designs, in simulation, an exploration policy whose real trajectory is most informative about physical parameters [22]. The work closest to mine uses a twin to find failure-prone configurations for real rollouts [13]. These works use the selected real data to correct either the model or the simulation. To my knowledge, correcting both from the same selected data has not been studied.

Finally, the gap can be decomposed into factors of variation whose difficulty ordering is largely consistent between simulation and reality [23]. It is rarely tested whether measurements taken before a method is applied predict, better than simple indicators such as the size of the gap, whether its benefit carries over to a new scene or to another task. My research aims to address these open questions.

(Please see section 12 for citations used in this document)

## 7. Pytania i hipotezy badawcze / Research questions and hypotheses

*Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).*

The aim of my dissertation is to develop representation learning methods that improve the generalization of deep learning models trained in digital twins built from real data, so that the models work reliably in reality and in places not seen during training. The methods should not be tied to a particular robot, scene or task.

I will not have a robot of my own at the start, and I will train models on academic GPU clusters, so the methods must work at this scale. I will rely on public datasets that contain independent captures of the same real scenes. I will study two tasks, robot navigation and robotic manipulation, and all results must be measured on held-out scenes that were not used for training or model selection.

My research questions and hypotheses are as follows:

1. Which reconstruction errors of a digital twin, varying across a scene, harm the generalization of navigation and manipulation models trained in it?
2. That the harm caused by an error in a region is predicted better by how much it changes the features of a fixed pretrained visual encoder on views rendered with and without the error than by its size in the image or geometry, e.g., PSNR, LPIPS or depth error. And that randomizing each region according to its estimated reconstruction error and task relevance generalizes better than uniform randomization of equal total strength.
3. At which stage of a model trained or fine-tuned in a twin do real inputs first lose task information that is available for simulated inputs? I expect an identifiable earliest stage, or a narrow range of stages, found by linear probes checked against control tasks, and I will test with causal interventions whether the model uses this information.
4. That aligning simulated and real representations at that stage reduces the simulation-to-reality gap more than the same alignment at the input, at the final features, at all stages or at layers chosen by existing criteria, under equal real data and training effort.



5. Which real data, chosen under a fixed budget, best correct both the digital twin and the model? I expect that using the same selected data to correct both reduces the gap more than correcting either one alone, and I will compare my selection with random and failure-driven selection.
6. Do the gains transfer, with unchanged settings, to unseen scenes and from one task to the other, and can properties of the shift measured beforehand predict this better than simple indicators, such as the size of the gap or the image-level discrepancy?
7. That, if the harm follows the size of the error rather than the change in features, reconstruction uncertainty and action sensitivity could still guide randomization and data selection, although uncertainty misses errors that the reconstruction reproduces consistently; I will follow this route mainly if the representation-based signals turn out to be uninformative.

## 8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

*Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).*

The main expected contribution of my dissertation is a set of representation learning methods that make deep learning models trained in digital twins generalize better to reality and to unseen places. The work will also show whether the simulation-to-reality gap is caused mainly by a limited set of task-relevant discrepancies, which can be found among the reconstruction errors of a twin, located inside the model and reduced with a small amount of well-chosen real data.

The work should provide evidence about which reconstruction errors harm generalization, at which stage of a model task information is lost on real inputs, when correcting both the twin and the model with the same real data pays off, and whether such gains carry over to new scenes and between navigation and manipulation. Because transfer from simulation to reality is a case of learning under distribution shift, these findings may also be relevant to domain adaptation, representation learning and active learning in general. I plan to release the code, the evaluation protocol and, where licences allow, the trained models, so that others can compare methods on held-out scenes in the same way. My work should be a scientific contribution bridging artificial intelligence and robotics, with practical significance for deploying learning-based robots in new environments such as warehouses, hospitals and homes.

## 9. Planowane metody badawcze / Planned research methods

*Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).*

My work will focus on building digital twins of real scenes, training and fine-tuning deep learning models in them, and designing methods that analyse and align the representations these models learn. Since no established theory guarantees generalization from simulation to reality, most of the work will be empirical. I will use domain adaptation theory as motivation and analyse the properties of my methods theoretically where possible.

I will use tools commonly employed in machine learning and robot learning. This includes PyTorch for model training, gsplat for 3D Gaussian Splatting [10] and COLMAP for camera poses, the Habitat simulator for navigation and ManiSkill3 for manipulation, Git with GitHub repositories for version control, and Hugging Face for retrieving and publishing models and datasets. I will start from



pretrained models where possible. I intend to work in the open and release code and evaluation scripts, and, where licences allow, trained models and derived data.

I will not have a robot of my own at the start, so the evaluation will use public data, and I will state this openly. For navigation, I will use public datasets of real indoor scenes with independent captures of the same scenes, such as ScanNet++ and MuSHRoom. I will build the twin from one capture, and the other will provide real frames and a dataset-based reference. This reference is a proxy for reality. It measures the gap caused by reconstruction fidelity and leaves out differences in actuation and sensors. For manipulation, I will build twins of real tabletop scenes from images of public robot datasets, following the visual matching of SIMPLER [14], and compare them with published real-robot results of the same policies. If access to a robotics laboratory of the university allows, I will use a real robot to check the direction of selected results and the validity of the proxy.

For the first research question, I will correct one type of reconstruction error at a time in selected regions using a more accurate reference, such as the laser scans of ScanNet++, or inject it in a controlled way, and measure the effect on the trained model. For the second question, I will train linear probes with control tasks on simulated hidden states and test them on paired real ones. Causal interventions on hidden states, such as patching the probed subspace from simulated into real inputs, will test whether the model uses the lost information. For the third question, I will compare rules for selecting real data with random and failure-driven selection at equal budgets, and compare correcting both the twin and the model with correcting only one. For the fourth question, I will apply the methods with unchanged settings to held-out scenes and test whether measurements taken beforehand predict the gains, comparing rank correlations with tests for dependent correlations.

I will measure the success rate and the simulation-to-reality gap, that is, the difference between the performance of the same model in the twin and in reality or its reference, on held-out scenes. I will compare methods with relevant baselines under comparable training effort and real data. To obtain reliable results, I will repeat runs over random seeds and scenes, use statistical tests for comparisons and run ablation studies. Data for training, model selection and final evaluation will be kept separate.

I will run experiments on the supercomputers of the Wrocław Centre for Networking and Supercomputing (WCSS), using their SLURM installation for job management, and on PLGrid resources. I intend to use AI coding assistants, such as Claude Code, to accelerate writing and testing code and to help with routine tasks, such as monitoring long experiments and proofreading.

I plan to publish my results at conferences and in journals from the ministerial list, such as ICRA, IROS, RSS and IEEE RA-L in robotics and NeurIPS, ICML, ICLR and CVPR in machine learning and computer vision. When the rules allow, I will also publish preprints on arXiv.

## 10. Streszczenie popularnonaukowe / Abstract for general public

*Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).*

### Streszczenie popularnonaukowe

Roboty coraz częściej uczą się w cyfrowych bliźniakach, czyli wirtualnych kopiach rzeczywistych miejsc odtworzonych na podstawie ich nagrań, ponieważ nauka w prawdziwym świecie jest powolna, kosztowna i czasem niebezpieczna. Taka kopia nigdy nie jest idealna. Na przykład szklane drzwi mogą zostać odtworzone błędnie i zmylić robota. W efekcie model, który dobrze działa w symulacji, w rzeczywistości często zawodzi, zwłaszcza w miejscach, których wcześniej nie widział. Chcę ustalić,



które różnice między kopią a rzeczywistością naprawdę szkodzą modelowi, w którym miejscu wewnątrz modelu zamieniają się w błędy oraz które nieliczne rzeczywiste przykłady najlepiej poprawiają zarówno model, jak i kopię. Sprawdzę to w nawigacji robotów oraz w chwytaniu i przedstawianiu przedmiotów. Oczekiwanym wynikiem są metody, dzięki którym roboty będą mogły szybciej, taniej i bezpieczniej uczyć się pracy w nowych miejscach.

### Abstract for general public

Robots are increasingly trained in digital twins, that is, virtual copies of real places rebuilt from their recordings, because learning in the real world is slow, costly and sometimes unsafe. Such a copy is never perfect. For example, a glass door may be rebuilt incorrectly and mislead the robot. As a result, a model that works well in simulation often fails in reality, especially in places it has not seen before. I want to find out which differences between the copy and reality actually harm the model, where inside the model they turn into errors, and which few real examples best correct both the model and the copy. I will test this in robot navigation and in grasping and moving objects. The expected result is methods that let robots learn to work in new places faster, more cheaply and more safely.

## 11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

*Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.*

September 2027 (planned submission to a robotics conference or journal from the ministerial list, e.g. ICRA 2028 or IEEE Robotics and Automation Letters).

## 12. Inne / Other comments

*Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).*

I prepared this plan with the help of AI tools (Claude, used through Claude Code). They were used for support in searching and checking the literature, drafting, proofreading and editorial support, and formatting assistance. I have reviewed all content, and the references were checked against publisher and preprint records.

### Bibliography

- [1] Kadian, A., et al. (2020). Sim2Real predictivity: Does evaluation in simulation predict real-world performance? RA-L.
- [2] Ben-David, S., et al. (2010). A theory of learning from different domains. Machine Learning.



- [3] Tobin, J., et al. (2017). Domain randomization for transferring deep neural networks from simulation to the real world. IROS.
- [4] Chebotar, Y., et al. (2019). Closing the sim-to-real loop: Adapting simulation randomization with real world experience. ICRA.
- [5] Tiboni, G., et al. (2024). Domain randomization via entropy maximization. ICLR.
- [6] Ganin, Y., et al. (2016). Domain-adversarial training of neural networks. JMLR.
- [7] Cheng, S., et al. (2025). Generalizable domain adaptation for sim-and-real policy co-training. NeurIPS.
- [8] Zhao, H., et al. (2019). On learning invariant representations for domain adaptation. ICML.
- [9] Lei, Y., et al. (2026). A mechanistic analysis of sim-and-real co-training in generative robot policies. arXiv preprint arXiv:2604.13645.
- [10] Kerbl, B., et al. (2023). 3D Gaussian splatting for real-time radiance field rendering. ACM TOG.
- [11] Torne, M., et al. (2024). Reconciling reality through simulation: A real-to-sim-to-real approach for robust manipulation. RSS.
- [12] Qureshi, M. N., et al. (2025). SplatSim: Zero-shot Sim2Real transfer of RGB manipulation policies using Gaussian splatting. ICRA.
- [13] Xu, Q., et al. (2026). TwinRL-VLA: Digital twin-driven reinforcement learning for real-world robotic manipulation. ACM MM (arXiv:2602.09023).
- [14] Li, X., et al. (2024). Evaluating real-world robot manipulation policies in simulation. CoRL.
- [15] Chhablani, G., et al. (2025). EmbodiedSplat: Personalized real-to-sim-to-real navigation with Gaussian splats from a mobile device. ICCV.
- [16] Xie, Z., et al. (2025). Vid2Sim: Realistic and interactive simulation from video for urban navigation. CVPR.
- [17] Jin, R., et al. (2026). Grounding sim-to-real generalization in robotic manipulation: An empirical study with vision-language-action models. ECCV.
- [18] Wang, X., et al. (2026). ReVeal: A reconstruction-aware real-to-sim framework for VLA policy evaluation. arXiv preprint arXiv:2609.23910.
- [19] Kachaev, N., et al. (2026). Don't blind your VLA: Aligning visual representations for OOD generalization. AAMAS.
- [20] Lee, Y., et al. (2023). Surgical fine-tuning improves adaptation to distribution shifts. ICLR.
- [21] Maddukuri, A., et al. (2025). Sim-and-real co-training: A simple recipe for vision-based robotic manipulation. RSS.
- [22] Memmel, M., et al. (2024). ASID: Active exploration for system identification in robotic manipulation. ICLR.
- [23] Xie, A., et al. (2024). Decomposing the generalization gap in imitation learning for visual robotic manipulation. ICRA.

### 13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

*Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.*

My supervisor and I plan weekly meetings, in which I present my progress and recent experimental results, we agree on the next steps and we work on joint papers. In addition, I will attend the weekly meetings of the supervisor's group of PhD students, where current results, problems and new work in the field are discussed. Day-to-day contact will be by email and instant messaging, and code, trained models and the LaTeX sources of papers will be kept in shared Git repositories.

### 14. Opinia promotora pomocniczego / Opinion of assistant supervisor



*Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.*

nie dotyczy / not applicable

.....  
Podpis promotora pomocniczego /  
Signature of assistant supervisor

## 15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....  
Data / Date

.....  
Podpis doktoranta / Signature of doctoral student

.....  
Podpis promotora / Signature of supervisor

.....  
Podpis drugiego promotora / Signature of second supervisor